

From Introduction to Advanced Deep Learning

Artificial Neural Networks · Convolutional Networks · Recurrent Networks · Detection · Segmentation & Tracking · Generative Models · Distillation & Deployment — each topic taught for four hours and closed by its own project(s), each with its own dataset and its own problem statement.

Professor	Dr. Muhammad Sajad	Contact hours	4 hours per topic · 7 topics · 28 h total
Presented by	Ahsan Umar — Instructor, Lab Instructor & Teaching Assistant	Projects	P1–P6 mandatory · P7–P12 optional
Stack	Python · PyTorch · torchvision · OpenCV	Track	Core = Topics 1–4 · Optional = Topics 5–7

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★ optional topic · ★★ advanced elective inside an optional topic. Mandatory core = Topics 1–4 (16 contact hours, 6 graded projects: P1–P2 for ANN, P3 for classification, P4 for detection, P5–P6 for RNN). Every topic closes with its own project(s) inside its four hours.

TOPIC 1 Artificial Neural Networks (ANN)

CORE

4 hours

Subtopics

- The perceptron and the artificial neuron; why depth + non-linearity (XOR)
- Activation functions: ReLU, LeakyReLU, GELU, softmax — when each fails
- Loss functions: MSE, cross-entropy, BCE; matching loss to output head
- Backpropagation and the computational graph; gradient checking
- Optimizers: SGD + momentum, Adam, AdamW; LR schedules and warm-up
- Regularization: dropout, weight decay, early stopping, augmentation; BatchNorm / LayerNorm
- Training-loop anatomy; under/over-fitting and vanishing-gradient diagnosis
- Training-loop anatomy checklist; reproducibility (seeds, splits, checkpoints)

Outcome: train, regularize and diagnose a fully connected network from scratch, and read every curve it produces.

Four-hour flow: h 0–1 neuron, activations, losses · h 1–2 backprop, computational graph, gradient checking · h 2–3 optimizers, schedules, regularization, norms, failure diagnosis · h 3–4 projects P1 + P2 (pair work, live code)

End-of-topic projects — one problem, one dataset, one deliverable each

Project	Problem / task	Dataset
P1 MANDATORY	Classify handwritten digits from raw 28×28 pixels with an MLP; establish the baseline accuracy and diagnose the loss curve (underfit / overfit / dying ReLU).	MNIST — 70,000 grayscale images, 10 classes (torchvision)
P2 MANDATORY	Regression instead of classification: predict a continuous target with feature scaling, MSE loss and R ² /MAE reporting — a different task shape on tabular data.	California Housing — 20,640 rows × 8 features (scikit-learn)

TOPIC 2 Convolutional Neural Networks — Image Classification

CORE

4 hours

Subtopics

- Convolution, stride, padding, pooling; output-size and receptive-field arithmetic; 1×1 and depthwise-separable convs
- Architecture ladder: LeNet → AlexNet → VGG → ResNet → EfficientNet
- Pre-training & transfer learning: frozen backbone, linear probe, fine-tune, head swap
- Classification pipeline: augmentation, splits, train/val discipline
- Metrics and reading a run: top-1 / top-5, per-class F1, confusion matrix, loss curves, early stopping

Outcome: build a convolutional classifier from scratch, then beat it with a transferred pre-trained backbone.

Four-hour flow: h 0–1 convolution arithmetic and the architecture ladder · h 1–2 transfer-learning regimes and the head swap · h 2–3 pipeline, augmentation, metrics · h 3–4 project P3

End-of-topic projects — one problem, one dataset, one deliverable each

Project	Problem / task	Dataset
P3 MANDATORY	Ten-class natural-image classification: train a small CNN from scratch, then beat it with an ImageNet-pretrained backbone + fine-tuning; report top-1 and the confusion matrix.	CIFAR-10 — 60,000 colour 32×32 images, 10 classes

TOPIC 3 Convolutional Networks — Object Detection

CORE

4 hours

Subtopics

- From classification to localization: bounding-box regression and delta encoding
- Detection as a variable-output problem: grids, region proposals, anchors
- IoU, label assignment, NMS; precision/recall, AP and mAP
- Two-stage family: R-CNN → Fast R-CNN → Faster R-CNN (RPN, RoI pooling)
- One-stage family: YOLO, SSD, RetinaNet (focal loss); the anchor-free idea
- Detector knobs and formats: confidence, NMS-IoU, input size, FPS; VOC/COCO boxes, YOLO txt labels

Outcome: extend a convolutional backbone into a working detector and tune it like a production system.

Four-hour flow: h 0–1 localization and box regression · h 1–2 IoU, anchors, assignment, NMS, AP/mAP · h 2–3 two-stage vs. one-stage families · h 3–4 project P4

End-of-topic projects — one problem, one dataset, one deliverable each

Project	Problem / task	Dataset
P4 MANDATORY	Multi-object detection: train a YOLO detector, sweep confidence × NMS-IoU against precision/recall, and measure real-time FPS — the classification backbone extended to boxes.	Pascal VOC 2012 — 17,125 images, 20 classes (or a 200-image custom YOLO-format set)

TOPIC 4

Recurrent Neural Networks & Sequence Models

CORE

4 hours

Subtopics

- Why sequences break feed-forward nets; the recurrent hidden state
- RNN cell, unrolling in time, BPTT
- Vanishing / exploding gradients; clipping and the five fixes
- LSTM: forget / input / output gates and the cell-state conveyor
- GRU; bidirectional and stacked RNNs; which to choose
- Variable-length batches: padding, packing, masking
- Embeddings and sequence heads (classification vs. per-step output)
- Seq2Seq encoder–decoder, teacher forcing, exposure bias
- The attention idea (concept only — no transformer layers are built)
- Where recurrence returns: tracking (Topic 4) and CRF-RNN segmentation

Outcome: model variable-length sequences and ship a sequence-to-sequence translator.

Four-hour flow: h 0–1 RNN cell, unrolling, BPTT, gradient pathologies · h 1–2 LSTM, GRU, bidirectional, packing and masking · h 2–3 embeddings, Seq2Seq, the attention idea · h 3–4 projects P5 + P6

End-of-topic projects — one problem, one dataset, one deliverable each

Project	Problem / task	Dataset
P5 MANDATORY	Machine translation: English→German with an encoder–decoder RNN + attention; score with BLEU and inspect alignment — the course's flagship sequence task.	Multi30k En-De — 29,000 train / 1,014 test sentence pairs
P6 MANDATORY	Variable-length sequence classification: embedding + BiLSTM sentiment classifier; compare plain RNN vs. LSTM on long reviews to expose the vanishing gradient.	IMDB movie reviews — 50,000 labelled reviews (binary sentiment)

TOPIC 5

Segmentation & Object Tracking

★ OPTIONAL

4 hours

Subtopics

- Semantic segmentation: FCN, transposed convolution, U-Net skips
- Dilated convolutions and DeepLab in brief; CRF-RNN link to Topic 3
- Instance segmentation: Mask R-CNN and RoIAlign; YoLACT idea
- Panoptic segmentation and the PQ metric (one slide)
- Dense-prediction losses and metrics: Dice, mIoU, mask-AP
- Single-object tracking and Siamese trackers
- Re-ID and similarity / triplet-loss embeddings
- Tracking-by-detection: Kalman filter + Hungarian assignment (SORT, DeepSORT)
- Tracking metrics: MOTA, IDF1, HOTA, ID switches
- Deployment link: run the detector every k-th frame, track in between

Outcome: move from boxes to pixels, and from per-frame detections to persistent identities.

Four-hour flow: h 0–1.5 semantic, instance and panoptic segmentation + metrics · h 1.5–3 SOT, Re-ID, tracking-by-detection (Kalman + Hungarian) · h 3–4 projects P7 + P8

End-of-topic projects — one problem, one dataset, one deliverable each

Project	Problem / task	Dataset
P7 OPTIONAL	Pixel-level semantic segmentation with a mini U-Net; ablate the skip connections and report the mIoU drop on real pet images.	Oxford-IIIT Pet — 7,349 images with pixel trimaps (or CamVid)
P8 OPTIONAL	Keep identities alive across frames: detector + Kalman + Hungarian; count ID switches with and without the motion model.	MOT17 — pedestrian sequences, train split (detections provided)

TOPIC 6

Generative Models: Autoencoder → VAE → Diffusion / Flow Matching

★ OPTIONAL

4 hours

Subtopics

- Discriminative vs. generative; the problem statement for $p(x)$
- Autoencoder: bottleneck, reconstruction loss, what it does and cannot do
- AE variants: denoising, sparse, contractive; anomaly detection by reconstruction error
- Why an AE cannot sample — the unstructured latent
- VAE: latent distribution, reparameterization trick, ELBO = reconstruction + KL
- What a VAE does: sampling, interpolation, latent arithmetic; β -VAE trade-off
- GANs in one pass: minimax game, mode collapse, where GANs still win
- ★★ Diffusion: forward noising, ϵ -prediction, iterative sampling, guidance
- ★★ Flow matching: velocity fields and straight-path ODEs vs. diffusion
- Generative models elsewhere: MAE pre-training, diffusion U-Nets, trajectory GANs

Outcome: explain what each generative family buys, and make a model sample, denoise and interpolate.

Four-hour flow: h 0–1 the autoencoder problem statement and variants · h 1–2 VAE: reparameterization, ELBO, what a VAE does · h 2–3 GAN overview; ★★ diffusion and flow matching · h 3–4 projects P9 + P10

End-of-topic projects — one problem, one dataset, one deliverable each

Project	Problem / task	Dataset
P9 OPTIONAL	Train an AE then upgrade the same code to a VAE: denoise corrupted digits, sample new digits from $z \sim N(0, I)$, interpolate latents, score anomalies by reconstruction error.	MNIST (corrupted) — synthetic noise / masking applied in code
P10 OPTIONAL ★★	Generate 32×32 images with a small diffusion model (or its flow-matching variant); compare sample quality and sampling cost against the P9 VAE.	CIFAR-10 (unsupervised use) or CelebA 64×64

TOPIC 7

Knowledge Distillation & Deployment

★ OPTIONAL

4 hours

Subtopics

- What distillation is: teacher, student, dark knowledge
- Temperature and the Hinton loss ($\alpha \cdot \text{CE} + (1-\alpha) \cdot T^2 \cdot \text{KL}$)
- Feature-based (FitNets, attention transfer) and relation-based KD
- Self-distillation, mutual distillation, data-free KD
- KD in computer vision: detector heads, segmenters, Re-ID embeddings
- Compression toolkit: INT8 PTQ / QAT, pruning, operator fusion
- Export formats: TorchScript, ONNX; runtimes: TensorRT, OpenVINO, TFLite
- Serving patterns: batch, online, real-time video, edge
- Latency percentiles, throughput, monitoring, drift and rollback
- The lifecycle loop: distill → quantize → export → benchmark → monitor

Outcome: compress a trained model without losing it, and ship it with numbers that survive production.

Four-hour flow: h 0–1 KD core: dark knowledge, temperature, Hinton loss · h 1–2 KD variants and CV-specific distillation · h 2–3 compression, export, serving, monitoring · h 3–4 projects P11 + P12

End-of-topic projects — one problem, one dataset, one deliverable each

Project	Problem / task	Dataset
P11 OPTIONAL	Distill a ResNet-50 teacher into a MobileNetV3 student; sweep temperature T and α ; plot the accuracy–latency Pareto against the undistilled student.	CIFAR-100 — teacher trained on it; student learns from soft targets
P12 OPTIONAL	Take the P4 detector through the deployment pipeline: ONNX export → INT8 quantization → benchmark p95 latency and mAP drift on held-out frames.	Pascal VOC 2012 val + a 300-image calibration subset

Reading the roadmap: each topic is four contact hours — subtopics first, then its two projects inside the same four hours. Mandatory track = Topics 1–4 with projects P1–P6 (ANN and RNN carry two projects each; classification and detection carry their own single project). Topics 5–7 (★) and their projects P7–P12 are optional and may be taken in any order after the core; inside Topic 6, diffusion and flow matching are ★★ advanced electives. Every project states its own problem and its own dataset, and no two projects in a topic share a dataset or a task type.